Sardar Patel Institute of Technology

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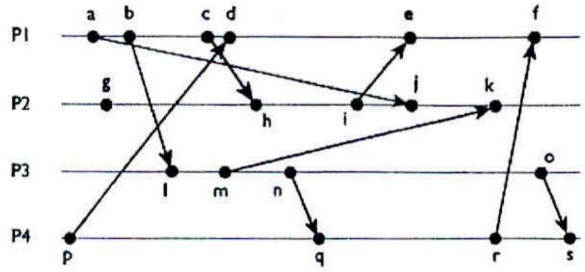
(Empowered Autonomous College of Affiliated to University of Mumbai)

End Semester Examination

December 2023

Max Marks: 100 Duration: 3 hours
Class: T.Y. B.Tech Semester: V
Course code:CS 304 Branch: Computer Engg.
Name of the course: Distributed Computing

Q No		Max Marks	CO	BL
Q.1 (a)	Describe Maekawa's Algorithm for distributed mutual exclusion, give an example with voting set n=4, and highlight its advantages and disadvantages over other non-token based algorithms.	10	3	3
Q.1 (b)	Explain different distributed computing models with the help of neat diagrams and give examples for the same.	10	1	4
Q. 2(a)	With the help of neat system diagrams, describe the following load balancing algorithms – 1. Round Robin with weighted servers 2. Least Response Time	10	2	3
Q 2(b)	Consider the behaviour of two machines in a distributed system. Both have clocks that are supposed to tick 1000 times per millisecond. One of them actually does, but the other ticks only 990 times per millisecond. If UTC updates come in once a minute, what is the maximum clock skew that will occur?	10	4	5
Q. 3(a)	What is the role of object adapters in object based distributed systems?	10	2	5

Q 3 (b)	Give the Lamport Clock timestamps for the following event diagram.  	10	4	5
Q.4(a)	With the help of timing diagram, describe the following consistency models – 1. Weak Consistency 2. Entry Consistency	10	3	5
Q.4(b)	What are two types of process addressing supported by a message passing system? and give the primitives for each addressing type. **OR** What are the three types of problems in IPC due to different forms of system failures, explain with the help of neat diagrams?	10	1	5
Q.5(a)	With the help of a neat diagram explain the principle of maintaining a migration stack to support migration of an execution segment in a heterogeneous environment.	10	3	3
Q.5(b)	Discuss epidemic protocols and describe its update propagation models and give its advantages.	10	3	3

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